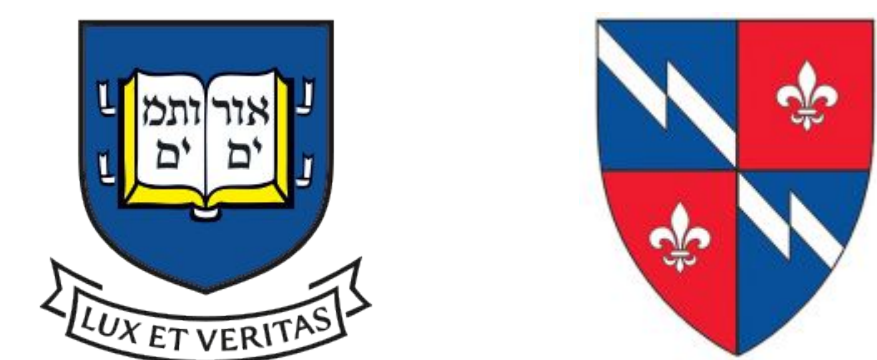
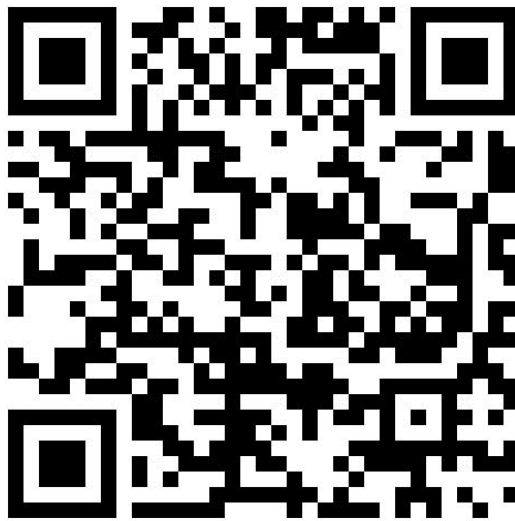




Developing a Player and Team Comparison & Analytics Tool for Yale Women's Soccer

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Shiny App

Abstract

The Yale Women's Soccer team competes in NCAA Division I and in the Ivy League Conference. The coaching staff would like to make decisions informed by data, but there is no qualified staff member who can turn the data into insights. Using the data available for all Ivy League teams and players, we built a Shiny app for various plots that can be used to compare any players from the Ivy League, and also analyze and compare Ivy League teams. Our platform gives an easy and accessible way to the coaching staff to use the data that is available without getting overwhelmed or distracted. Our aim for this platform is to help coaches with data-driven roster building and opponent scouting. The platform can be updated when new data is available.

Data Cleaning

- Combined individual player files (one row per game) into a single dataset covering the 2021–2024 seasons
- Calculated per 90 statistics for fair player comparison
- Excluded games where a player played fewer than 15 minutes
- Removed players with fewer than 90 total minutes played
- Assigned a primary position for each player per game and grouped positions into four categories: GK, DEF, MID, ATT
- Created percentage-based variables for selected metrics

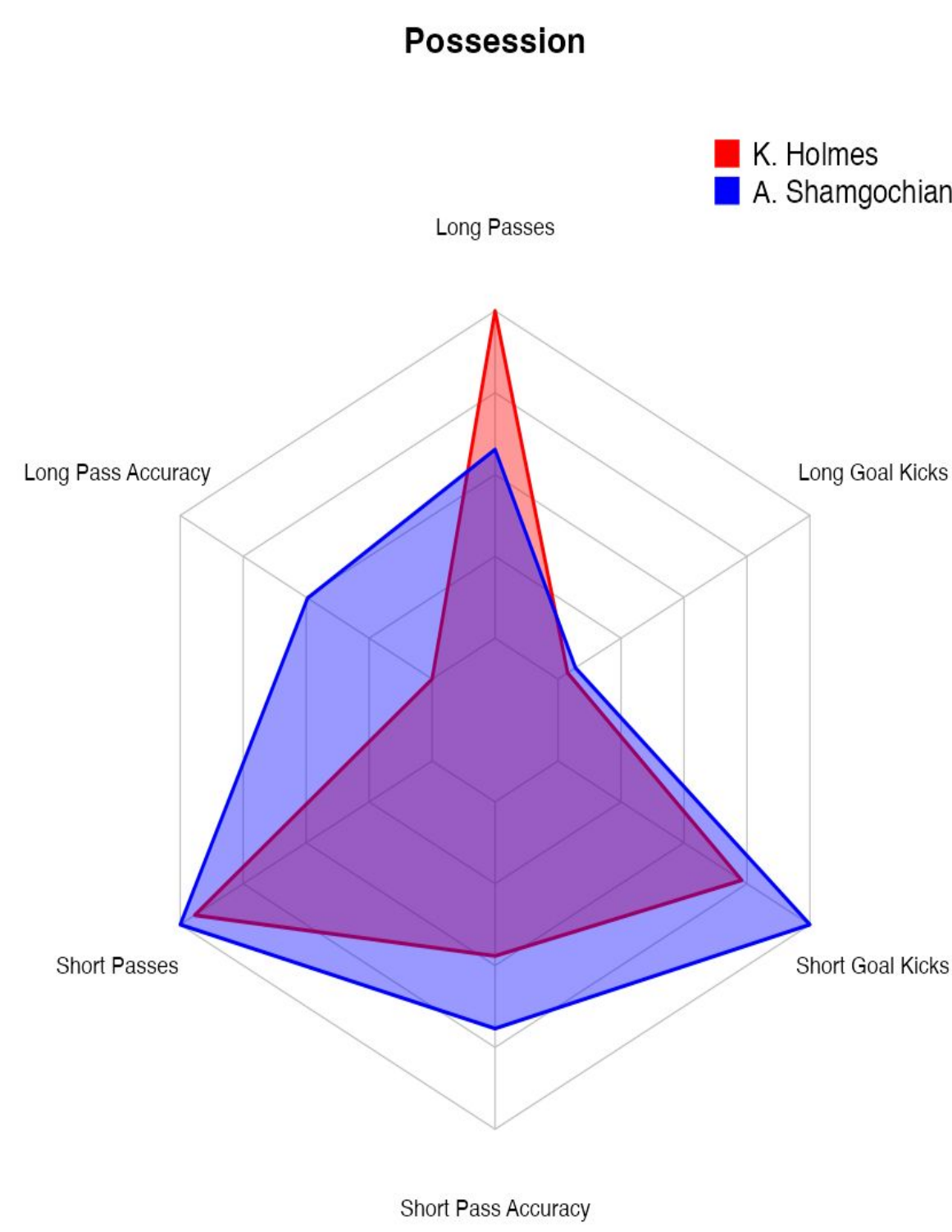
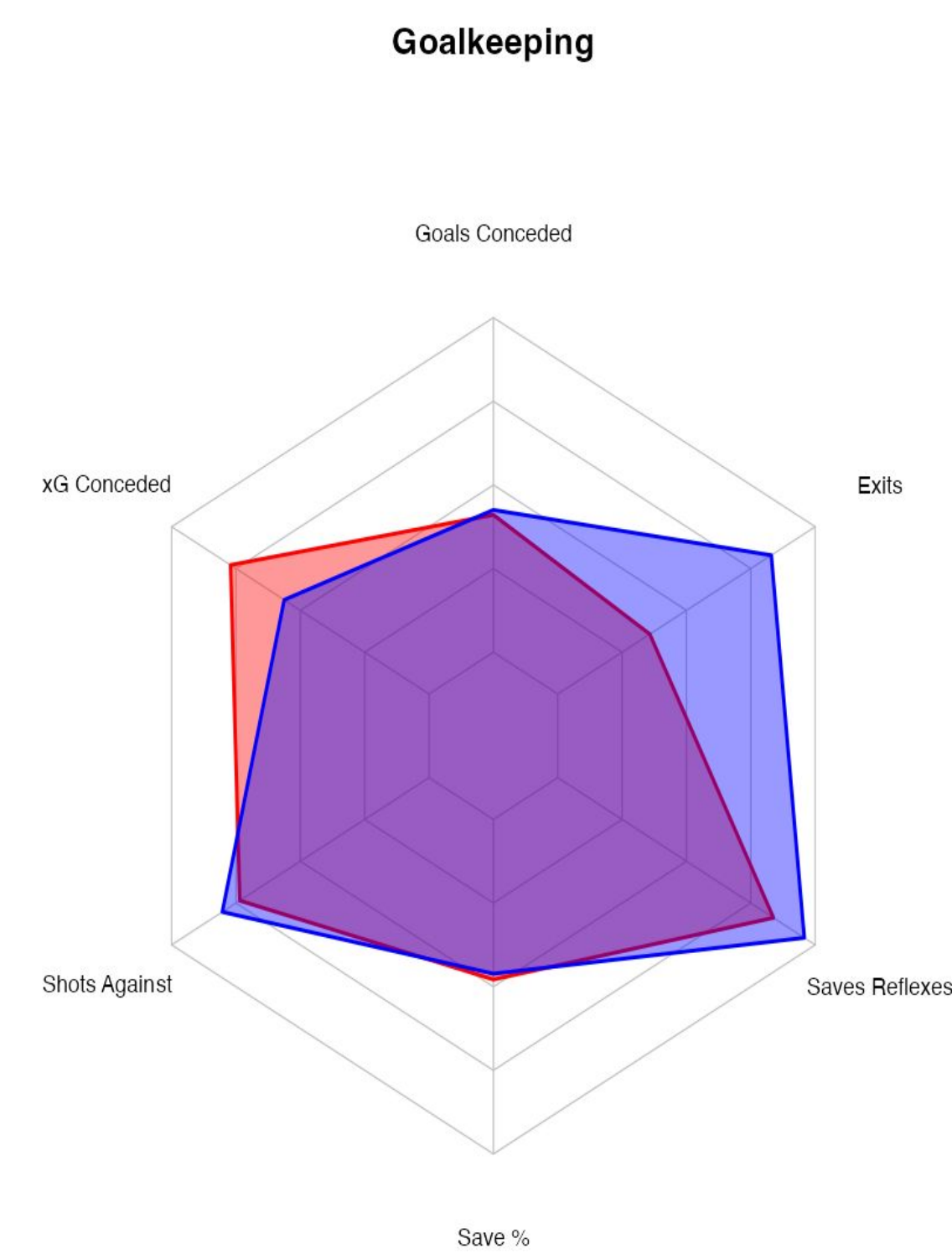
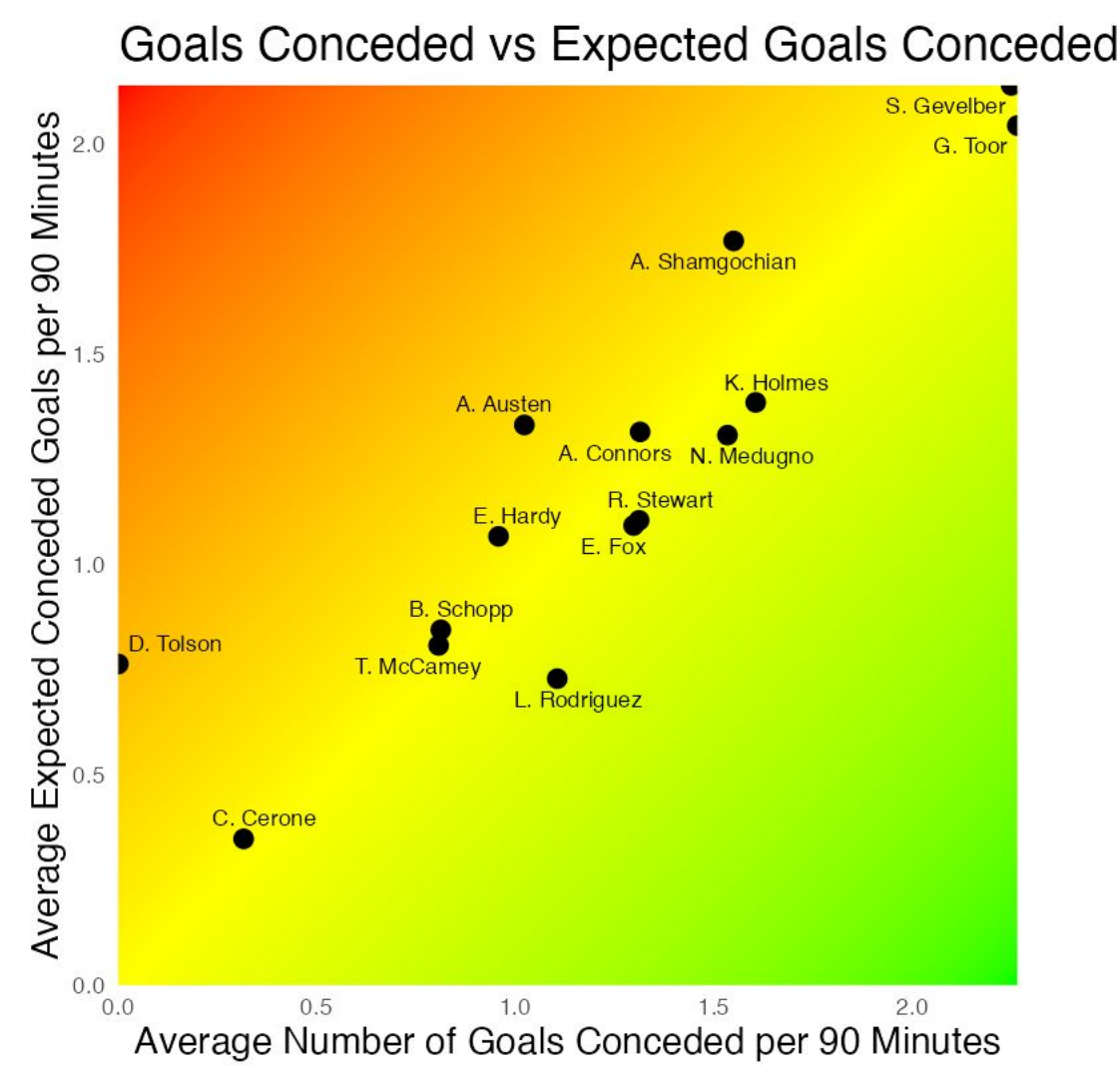
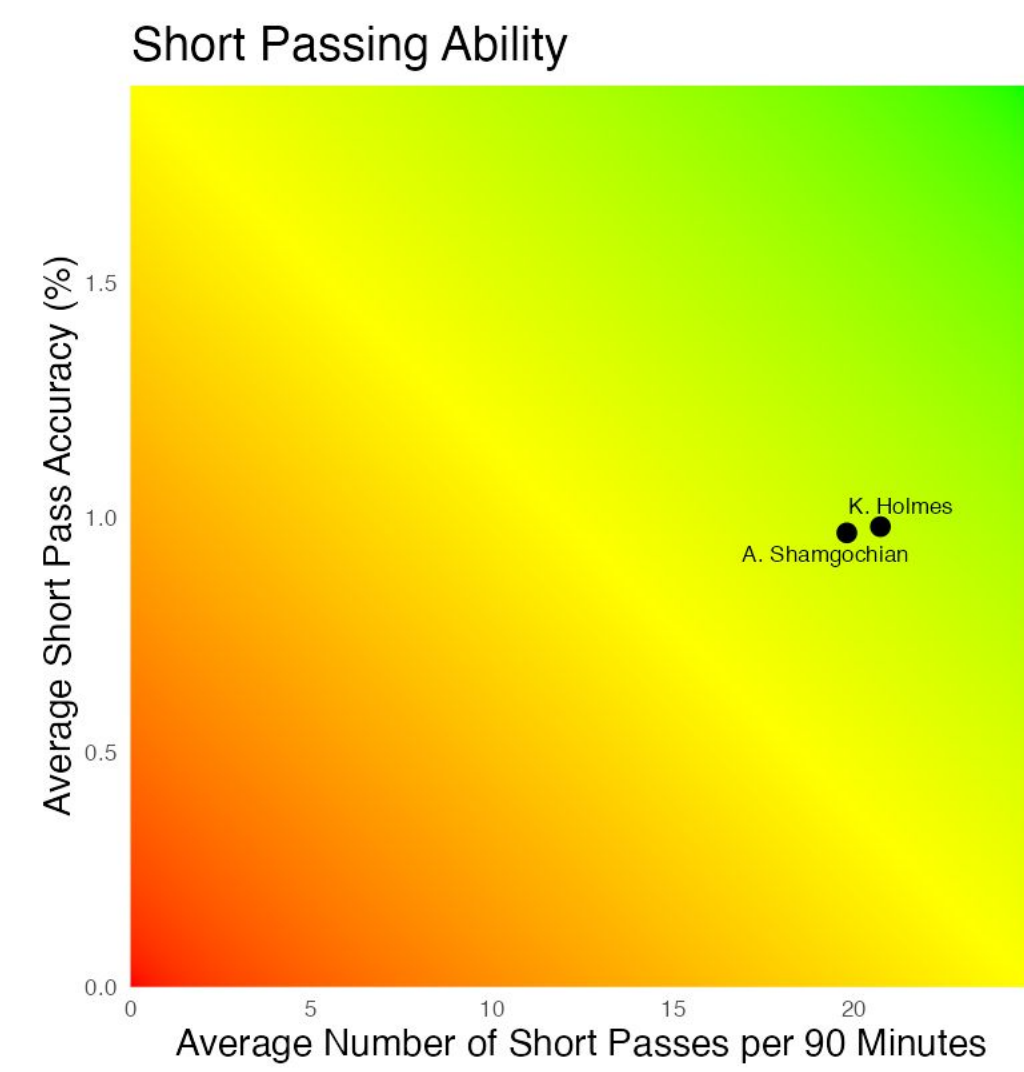
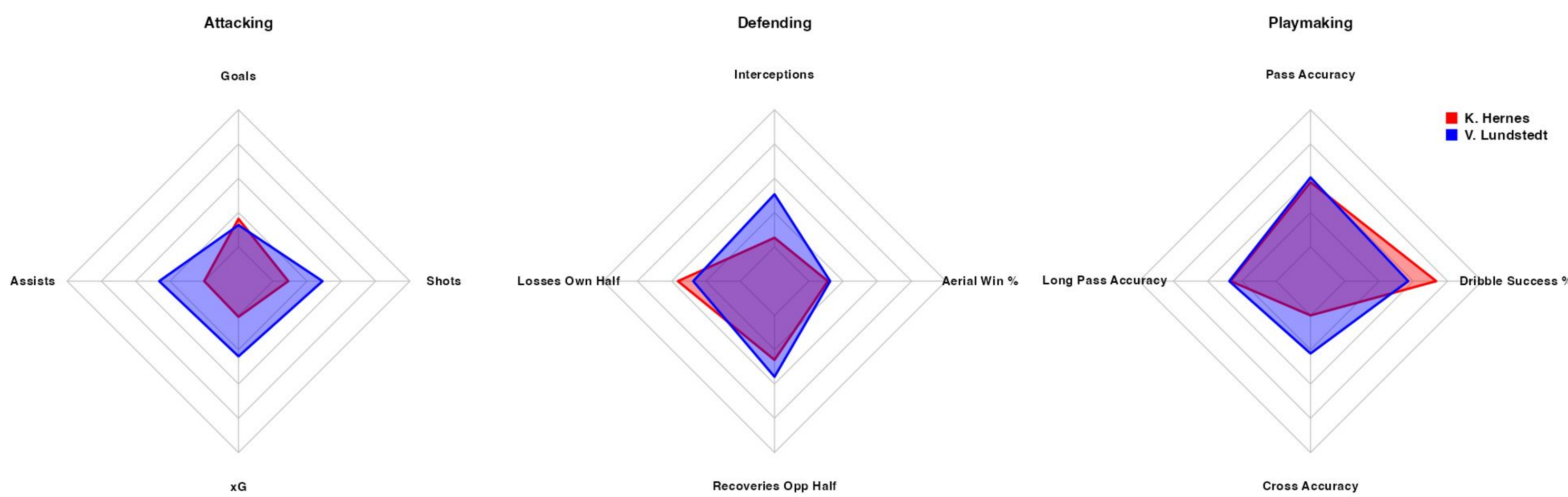
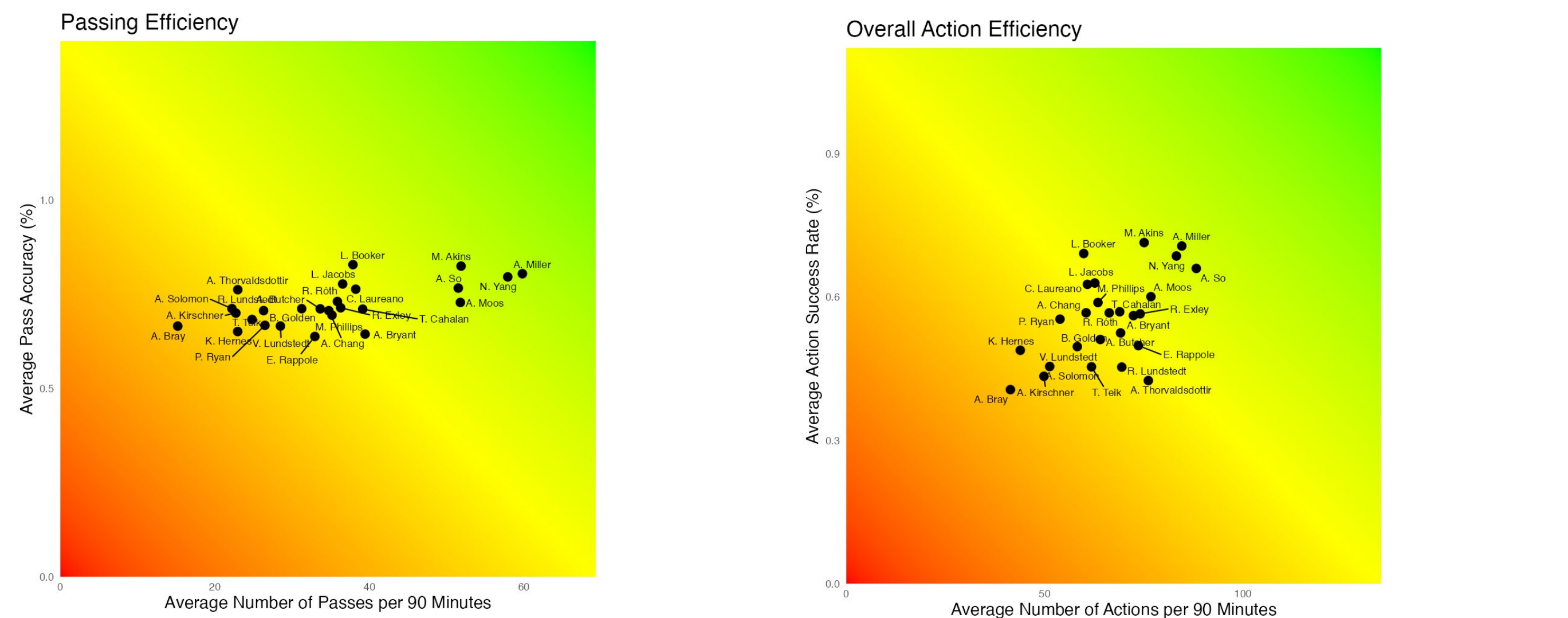
DATA STORYTELLING

- Yale has the ball the most, but has the second least shots on goal
- Yale loses the most challenges and gives up the third most xG
- Defenders have the most actions and the most passes in the Yale team
- Conclusion: Yale possesses the ball in their defensive unit rather than up front to create chances

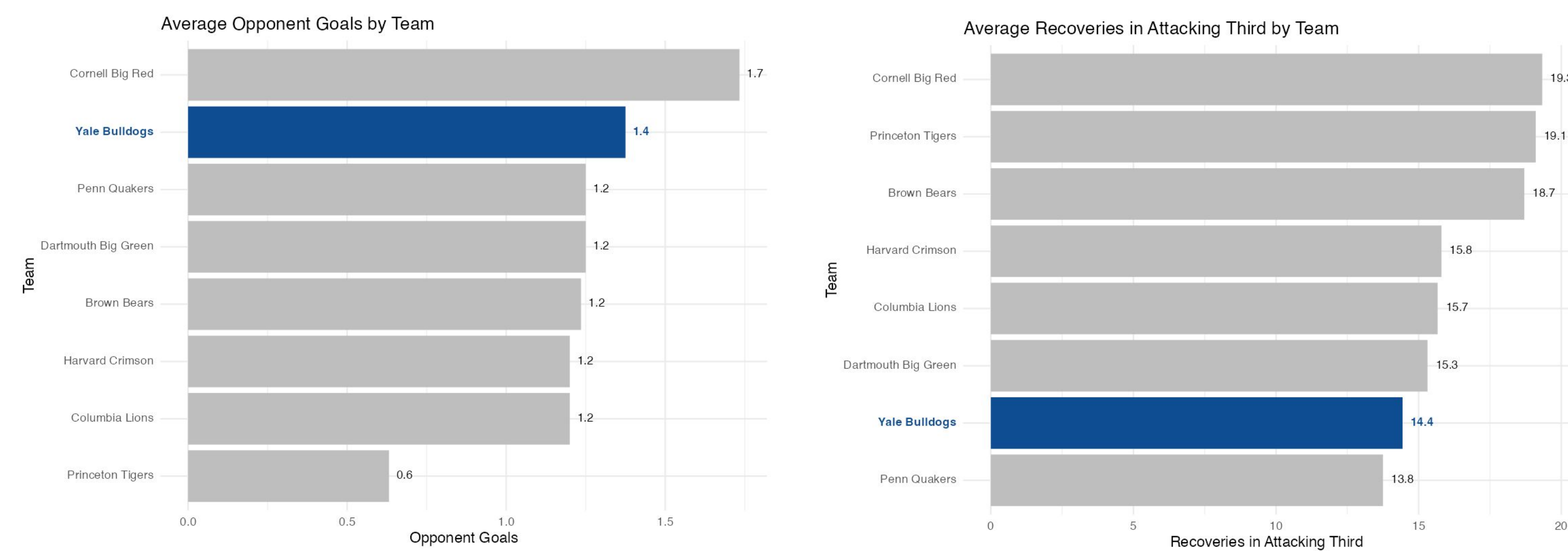
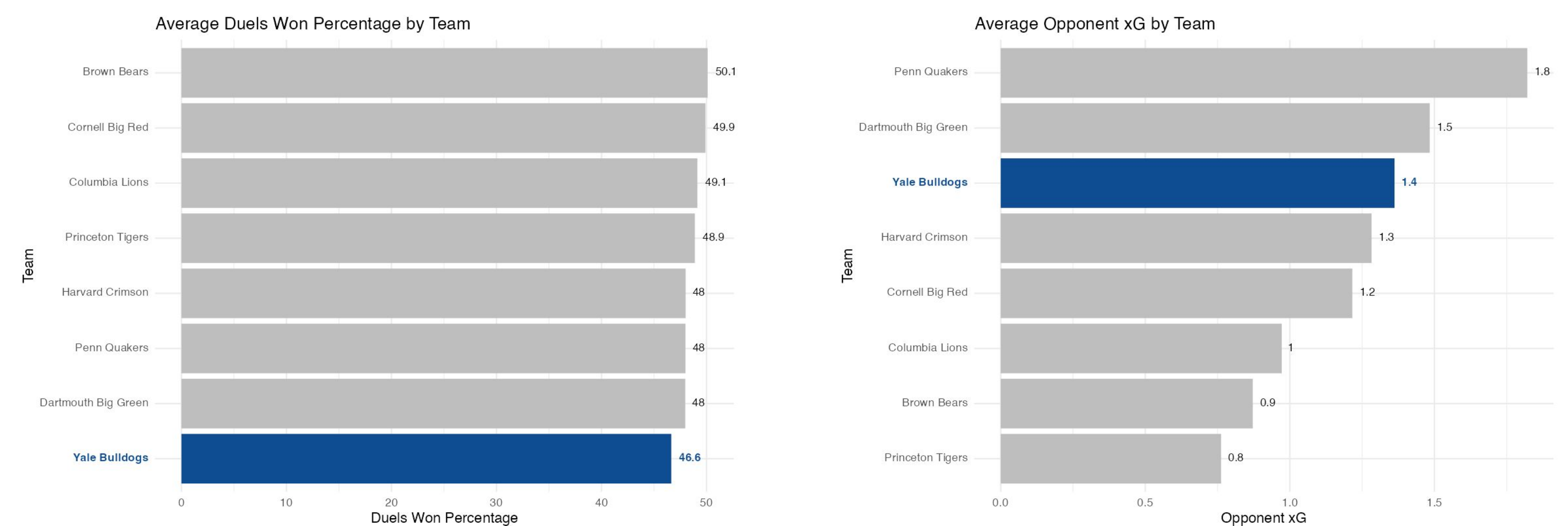
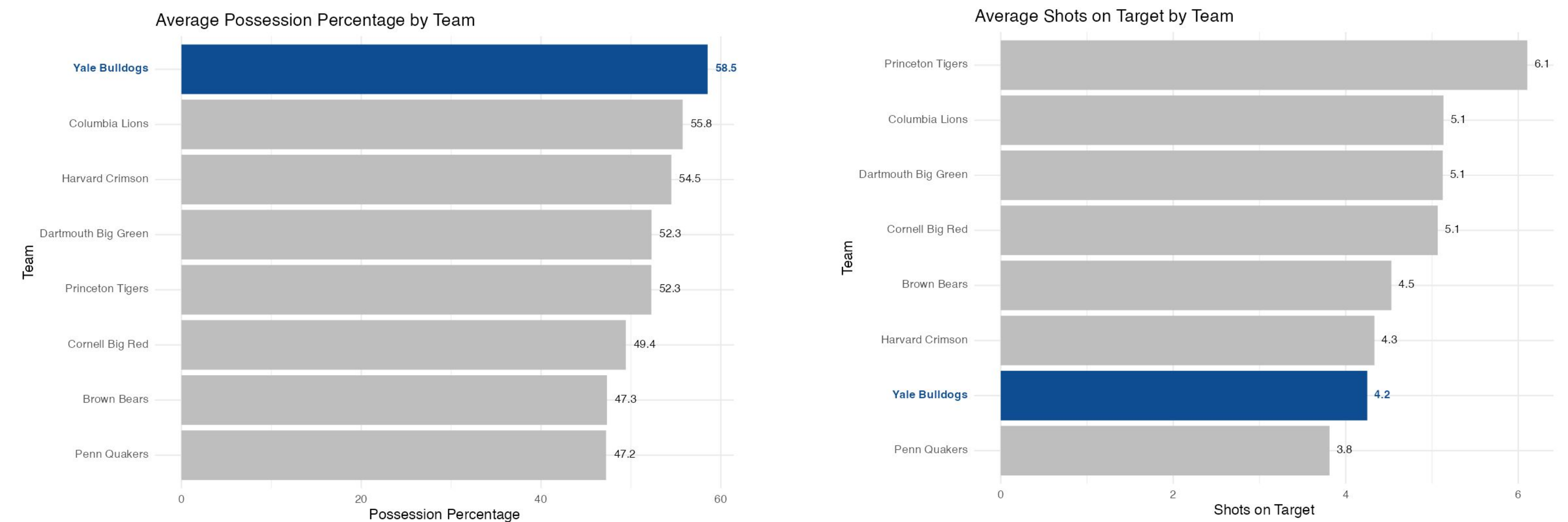
Team Comparison



Player Comparison



Team Analysis



Conclusion and recommendations

This project is part of a senior thesis and will be due in 3 weeks. The coaching staff will have access to the final product, the Shiny App. Whether it is used to pick the starting line-up against a given opponent or choose a tactic for the next game, we hope our tool will help the coaches make data-driven decisions that result in winning games. In the next 3 weeks we are planning to improve on the followings:

- Sit down with the coaches again and ask for their feedback
- Add more customization to radar charts (position, season, last n games)
- Add more customization to GK scatter plots (season, last n games)
- Build a model that predicts how well a team will do in the Ivy League based on their non-conference results
- Make Shiny design more professional and user friendly
- Add text to help user understand the plots